The MEW nozzle in two dimensions

Results at the corrected feed, for the set-up of Ju, Omar, Mirihanage and Kilic

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Summary. The set-up of *Suppressing oozing during filament-based melt electrowriting* (SSRN 7582986) was simulated as a planar slice through the nozzle: PCL at 120 °C fed through a 0.4 mm nozzle at 6.5 kV, 5.5 mm above a collector moving at 35 mm s^{-1}, at the corrected feed of 0.0208 mm^3 s^{-1} (Table 1 prints 0.255 mm^3 s^{-1}). Four results:

- With the field off, the melt hangs from the nozzle as a drop that grows at the feed rate and reaches the collector after 86 to 111 s at a nominal 500 Pa s, depending on the cell size and on the model of the melt's surface, and sooner at the measured 820 Pa s; nothing in a planar model can carry it away as a jet.
- With the field on, the result depends on how deep the model lets the melt conduct. The melt's estimated conductivity makes the whole drop a conductor; the field then pulls it into a column that reaches the collector 27 s before the field-off drop. Shallower conducting depths delay it instead, so this rests on that estimate.
- At 35 mm s^{-1} a straight fibre carrying the whole feed is 27.5 µm across; the reported 19 µm to 22 µm fibres carry about half of it, the fan's 9.5 µm to 13.5 µm fibres 12 to 24 %. Neither the Taylor cone nor the lag angle can account for the rest.
- Part of the dependence on the cell size came from how the model handled the melt's surface, and the surface tension on it, where it crosses the cells as a staircase. With that corrected, the field-off drop touches at 109, 96 and 86 s on 100, 50 and 25 µm cells: closing in, not yet converged. The field-on runs still use the earlier model, and nothing after contact with the collector can be used.

1 Model and set-up

Melt	PCL held at 120 °C. Viscosity measured: 780 to 850 Pa s, close to Newtonian from 0.4 to 16 s^{-1} (the bore flow is about 3 s^{-1}); runs at 820 Pa s and at a nominal 500 Pa s. Surface tension 0.03 N m^{-1} is nominal; density 1080 kg m^{-3}.
Nozzle	0.4 mm bore, as in the paper; an assumed 1.0 mm tip face, 5.5 mm above the collector.
Field	6.5 kV. The melt conducts down to a chosen depth below the tip (2 mm by default; 1, 3 and 5 mm tested) and is an insulator below it, so it cannot short the gap.
Collector	35 mm s^{-1} (2100 mm min^{-1}), as in every experiment.
Feed	0.0208 mm^3 s^{-1}, Table 1's corrected value (Murat Kilic, 3 November), with and without the fan; the melt leaves the bore at 0.17 mm s^{-1}. The note of 3 November has the two earlier readings.
Grid	25 µm cells over an 8×6.8 mm slice, 14 µs steps, with 50 and 100 µm cells as checks. A free-surface lattice Boltzmann model of the melt (D2Q9, central moments; the air is not simulated) and Laplace's equation for the potential in the air (D2Q5); the melt surface feels the pull $\varepsilon_0 E^2/2$.

Matched to the experiment: the capillary numbers of the feed (2.8 at 500 Pa s, 4.5 at 820 Pa s) and of the collector (583, 957), the Bond number (0.014), and the ratio of electric to capillary pressure, $\varepsilon_0 V^2/\gamma R = 62$ at 6.5 kV. Not matched: inertia, which is negligible in the experiment and kept small in the model (Reynolds number at most 0.32).

2 Results

2.1 Field off: a drop that reaches the collector in under two minutes

At 500 Pa s the capillary number in the bore is 2.8. The melt hangs from the nozzle as a drop that grows at the feed rate; its lowest point falls at about 0.04 mm s^{-1} and reaches the collector after 105.0 s on 25 µm cells and 110.3 s on 50 µm cells (Fig. 1), 22 times later than the printed feed's column; with the corrected surface model (Section 2.4) it touches sooner, at 86.4 and 96.3 s. The melt within 1 mm of the tip settles at 0.69 mm^2, about what the printed feed piles there. A planar sheet carrying this feed at 35 mm s^{-1} would be 1.9 µm thick, far below the cells, so nothing carries melt away and no jet forms. In three dimensions a drop of the same outline needs more melt, by about its width over the bore's, four to five times here, so a round nozzle would take longer to reach the collector: an estimate, not a run. At the measured 820 Pa s the capillary number is 4.5 and the melt comes out more like a column: on 50 µm cells it touches at 91.3 s, not 110.3 s.

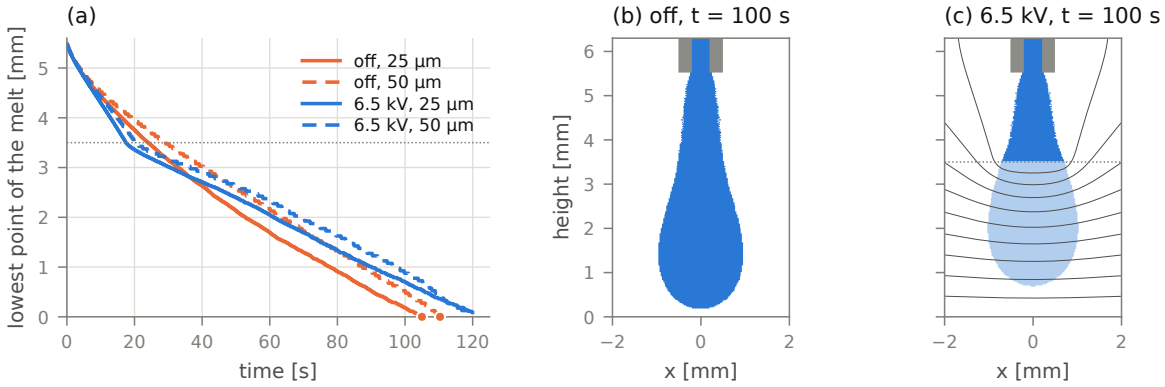


Figure 1. The corrected feed, 0.0208 $mm^3\ s^{-1}$, at 500 Pa s, earlier surface model. (a) Height of the lowest point of the melt against time with the field off (orange) and on (blue), on 25 µm (solid) and 50 µm cells (dashed). Dots mark first contact with the collector, which neither field-on run reaches by 120 s; dotted, the depth below which the melt does not conduct. (b, c) The drop at 100 s on 25 µm cells, field off and on; in (c) the melt that conducts is dark and the melt below the limit light, with equipotentials every 10 % of the voltage.

2.2 Field on: later or sooner, depending on a choice in the model

With the field on (these runs use the earlier surface model) the drop first falls faster, 0.40 mm (25 µm cells) and 0.45 mm (50 µm) below the field-off drop at 20 s. Then it slows and widens (2.05 to 2.10 mm at 100 s, against 1.90 mm) and is still 0.09 and 0.075 mm above the collector when the runs end at 120 s; extrapolated, it touches about 18 and 11 s after the field-off drop. The slowdown comes exactly when the drop's lowest point leaves the zone where the model lets the melt conduct, 2 mm below the tip: its speed falls there from 0.090 to 0.034 mm s^{-1} (50 µm), while the field-off drop passes the same height unchanged at 0.05 mm s^{-1}. The model needs that limit, or the melt would short the gap, but nothing in it fixes the depth. With the limit at 1 or 3 mm instead (Fig. 2a) the bend moves with it, the runs are identical to every printed digit until the drop first passes the shallower limit, and the drop reaches the collector 7.4, about 11 and 17.0 s after the field-off drop for 1, 2 and 3 mm.

While the whole drop conducts, its strongest pull is at its lowest point and draws it down. Once it hangs below the limit, its conducting part ends in a flat face inside it, and the strongest pull moves to that face's edge on the drop's sides, 7 to 56 times the typical pull on the conducting surface (Fig. 2b). That edge is numerical. With the limit at 5 mm, 0.5 mm above the collector, the whole drop conducts until it is nearly down, as a melt whose charge relaxes in much less than its two minutes of growth would. The field then draws it into a column that runs away toward the collector, 2.5 to 0.5 mm in 5.5 s, and it touches 48 s *before* the field-off drop, with 9 to 14 kV mm^{-1} in the last millimetre, well above the roughly 3 kV mm^{-1} at which air breaks down. The melt's conductivity was not measured, but its estimate, 10^{-11} to 10^{-4} S cm^{-1}, relaxes charge in at most about 0.03 s, far faster than the drop changes: the whole drop conducts, and the 5 mm case is the physical one. At 820 Pa s it touches at 64.7 s, 27 s before the field-off drop.

2.3 What the fibres carry

A straight fibre laid at 35 mm s^{-1} carries its cross-section times that speed; carrying the whole feed it would be 27.5 µm across. The paper's 19 µm to 22 µm fibres (after nozzle cleaning) carry 48 to 64 % of the feed, its 9.5 µm

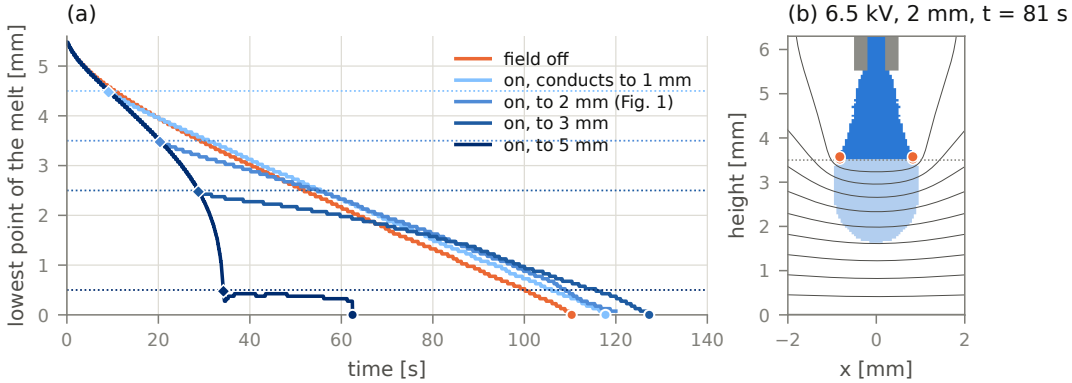


Figure 2. How deep the melt may conduct, on 50 µm cells, earlier surface model. (a) Height of the lowest point of the melt against time with the field off (orange) and on, the melt conducting to 1, 2, 3 or 5 mm below the tip (light to dark blue; 2 mm is the default of Fig. 1). Dotted, each depth; diamonds, the drop's lowest point passing it; dots, first contact. (b) The 2 mm run at 81 s: melt that conducts (dark) and melt that does not (light), equipotentials every 10 % of the voltage, and the four surface cells pulled hardest (orange), at the edge of the conducting zone.

to 13.5 µm fibres (with the fan) 12 to 24 %. A Taylor cone holds a fixed amount once formed, 0.3 to 5 s of feed, against a deficit of 0.45 to 1.1 mm^3 every minute. A lag angle thins the jet, but a fibre lying on the collector moves with it, so a 20.5 µm fibre carrying the whole feed would have to be laid at 63 mm s^{-1}, which coils it. At the same setting and speed the fan's fibres carry about a third of the melt of those without, and no oozing is seen, which suggests the drive delivers less than its setting with the fan on (question 1).

2.4 How the numbers depend on the cell size

Two things made the times move with the cell size. The model treats the nozzle as a surface the melt does not wet, so the melt creeps exactly two cells onto the tip face whatever a cell measures: the neck is 0.8, 0.6 and 0.5 mm wide on 100, 50 and 25 µm cells, and a narrower neck lets the drop fall faster. And the drop itself sagged more slowly on finer cells (a wetted tip, which fixes the neck, showed it). That effect is now traced and corrected: where the surface crosses the cells as a staircase, the model's bookkeeping of the melt in each surface cell left small ripples that the surface tension read as curvature, holding a relaxing drop back, more so on finer cells (Section 4 has the free-drop test). The field-off drop then touches at 109.4, 96.3 and 86.4 s on 100, 50 and 25 µm cells, against 111.2, 110.3 and 105.0 s before (Fig. 3): each halving now moves it about three quarters as far as the last (13.1, then 9.9 s), where before each moved it further. The times are closing in, too slowly for any to be final and more slowly than the neck alone explains (its excess over the bore halves each time); the rest is not yet identified. The field-on runs, those at 820 Pa s and the wetted tip used the earlier model. Nothing after contact can be used: the collector drags the drop to the box's side within 2 s.

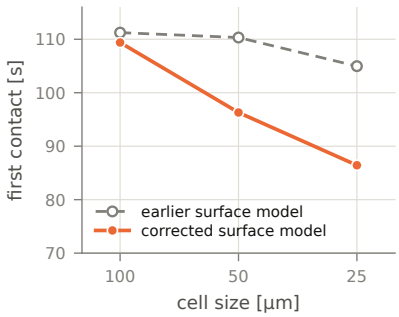


Figure 3. First contact of the field-off drop at 500 Pa s against the cell size.

3 What the model leaves out

Three dimensions: a planar meniscus has no conical equilibrium, the jet is far below the grid, a flat tip concentrates the field less than a round one, and a planar drop holds less melt for its outline. **Charge and current**: the melt is a perfect conductor down to a chosen depth and an insulator below. **Heat**: the melt is held at 120 °C; the hot end, where the fan acts, was modelled separately. The melt is Newtonian, which the measured curve supports at the nozzle's shear rates.

4 How far to trust it

Electric pull, against exact solutions	Its pattern is right to 0.05 %, a moving surface's response to 3 % once the surface's own stiffness is allowed for: deflections of one or two cells come out 9 % to 25 % short.
Surface tension, against exact solutions	At rest, Laplace's law to 1.2 % from 12 cells, 3.4 % at 8 (the bore's half-width); relaxing, a free drop's exact rate to 2 % from 32 cells, corrected model.
Cell size, 100, 50 and 25 µm	Not converged: field-off contact at 109, 96 and 86 s with the corrected model, 111, 110 and 105 s with the earlier one, which the other runs use.
How deep the melt conducts	Sets the field's effect: at 500 Pa s the drop reaches the collector 7.4, about 11 and 17.0 s after the field-off drop for depths of 1, 2 and 3 mm, and 48 s before it for 5 mm, the case the estimated conductivity points to.
Mass	Conserved to 0.4 % until contact (0.6 % at the printed feed); after contact off by up to 3.3 %, and by up to 5.4 % while a thread is drawn thin at the printed feed, which is not yet explained.

5 Questions for the experiment

1. How much filament is drawn in over 10 min, with the fan on and off? At the nominal feed it is 5.2 mm; the fibres imply about half that without the fan and a fifth with it.
2. Could the conductivity be measured, even to an order of magnitude? It would confirm that the whole drop conducts.
3. Is there a surface tension for this PCL near 120 °C? The model uses 0.03 N m^{-1}.
4. With the voltage off, how long does the melt take to reach the collector? The planar model gives about one and a half minutes, less on finer cells; a round nozzle would take longer.

M3LB (github.com/alexderosis/M3LB, branch `electrowriting`): `demonstrator/mew_nozzle -feed 0.0208` with `-nofield`, `-mu 820`, `-lc 1`, 3 or 5, `-film 0.1 -wet 60` (wetted tip), `-hf -geom -donor -follow` (corrected surface); cells `-R 8`, `-R 4 -Ucol 0.04`, `-R 2 -Ucol 0.08`.